Limitations of first law of thermodynamics

LIMITATION 1-
Does not indicate the direction of heat transfer

- The First Law cannot predict the direction in which heat will flow. For example, it cannot explain why heat flows naturally from a hot body to a cold body and not in the reverse direction without external assistance.

LIMITATION 2-
Does not specify the feasibility of a process

- It cannot determine whether a thermodynamic process is possible or impossible. A process may satisfy the energy conservation principle but still be physically impossible.

LIMITATION 3-
Does not give the maximum conversion of heat into work

- The First Law does not specify the maximum amount of heat that can be converted into useful work by a heat engine. In practice, no cyclic heat engine can convert all the heat supplied into work.

LIMITATION 4-
Does not explain why heat cannot be completely converted into work in a cyclic process

- According to the First Law, energy is conserved. However, it does not explain why a heat engine must reject some heat to a lower-temperature reservoir during a cycle.

LIMITATION 5-
Does not account for the quality of energy

- The First Law considers the quantity of energy but not its quality or usefulness. For example, electrical energy can be converted almost entirely into heat, whereas converting heat completely into work in a cyclic process is impossible.

LIMITATION 6-
Does not explain irreversibility

- The First Law does not distinguish between reversible and irreversible processes. It cannot explain why friction, free expansion, and heat transfer across a finite temperature difference are irreversible.